

Model Evaluation Plan (Capstone Project)

This document outlines the comprehensive evaluation framework used in the Stock AI Robo-Advisor project, integrating time-series models and fundamental financial data extracted from company reports.

Data Sources

- 1) **Stock Price Data**- Daily "OHLCV" data (Open, High, Low, Close, Volume) is taken from Yahoo Finances
- 2) **Financial Reports**- Quarterly company data (Revenue, Net Profit, ROE, EPS, etc.) are extracted from both TASE'S MAYA interface and Yahoo Finance API.

Stock EDA & Feature Engineering

Before modeling, a structured EDA process is applied for each ticker:

- **Missing Handling**: Failed tickers are logged; NaN imputation via `.bfill()` and `.ffill()`.
- **Feature Engineering**:
 - Rolling means: 5d, 10d, 20d of adjusted close.
 - Volatility: 10-day standard deviation.
 - Differences: High-Low and Open-Close.
 - RSI and MACD technical indicators.
 - Binary up/down movement flags.
 - Returns (percentage daily change).
- **Data Aggregation**:
 - Per-ticker groupings
 - Concatenation to a multi-index structure.

Feature Extraction Pipeline

1) reports to json files:

- Parses quarterly financial reports sourced from the TASE MAYA system.
- Utilizes string pattern matching & Regex to extract structured financial metrics from unstructured reports.
- Fields extracted include: Total Revenue, Net Profit, EPS, Total Assets, Equity, Operating Profit, and Cash Flow.
- Extracted values are normalized (e.g., converting "אלפי ש"ח" to numerical values in millions).
- Each extracted entry is labeled with company name, report date, and the corresponding quarter (YYYY-QX format, by the json file name)

2) Json semi-structured data to csv structured data:

- Links raw extracted metrics to official company tickers via handy-made mapping from .xlsx. file

- Produces a consolidated per-quarter dataset ready to be merged with time-series features for modeling.

3) Correlations between following quarters:

- For a given company, computes Pearson correlation between each financial parameter and the company's stock returns over future quarters (lags of 1–4).
- Visualizing correlation heatmaps and identifies company-specific financial drivers (e.g., EPS strongly predicting T+2 price for ACCL.TA).
- Allowing dynamic filtering of features by correlation threshold for personalized feature selection per ticker.

Modeling Framework

Each stock is modeled independently using the following configurations. All notebooks follow the structure of: **data preprocessing** → **model definition** → **training loop** → **evaluation**.

LSTM & GRU (Deep Learning)

- Input: sequences of 32 days, optionally concatenated with aligned report parameters.
- Architecture:
 - 4 recurrent layers (100 units)
 - 3 dense layers (25 → 25 → 1)
 - Dropout: 10%
- Optimizer: Adam
- Loss: Custom `volatility_loss` = MSE + volatility mismatch penalty
- Epochs: 30
- Evaluation split: last 10% of timeline
- Scaling: RobustScaler or MinMaxScaler
- Forecasting: 30-day forward simulation using normal-sampled percentage changes
- Interpretability: LIME used post-hoc on GRU/LSTM predictions

LightGBM & XGBoost (Boosted Trees)

- Input: Tabular features
 - Technical indicators (OHLCV-based)
 - Financial report metrics (aligned to time index)
- Parameters:
 - `n_estimators`: 100
 - `max_depth`: 6
 - `learning_rate`: 0.05
 - `objective`: regression
- Evaluation on last 10% of sorted data
- Feature importance extracted for interpretation
- Forecasting: 30-day forward simulation using normal-sampled percentage changes

Model Evaluation Metrics

- **Δ MAPE**: Measures the improvement (or lack thereof) when including financial report features versus time-series-only models. **The MAPE difference will be our main metric differing the experiment results, answering the most important question- Are report parameters contributing to stocks prediction?**
- **MAPE (Mean Absolute Percentage Error)**: Used as the primary metric for evaluating relative forecasting accuracy, especially useful for financial data due to its scale-invariance.
- **R^2 (Coefficient of Determination)**: Measures how well the predicted values explain the variance of the actual target values. Useful for detecting underfitting.
- **MAE / MSE (Mean Absolute Error / Mean Squared Error)**: Provide robust support metrics to evaluate prediction error magnitude, with MSE penalizing larger errors more heavily.
- **Negative R^2 Count**: Tracks how many stocks had $R^2 < 0$, indicating models that failed to capture even baseline variance.

Experimental Settings

- Two experimental conditions:
 1. **Baseline**: OHLCV time-series only.
 2. **Extended**: OHLCV + report parameters.
- Each model is trained per condition for all available stocks.
- Per-ticker selection of best-performing model (lowest MAPE).

Output Summary

Summarizing our experiments, with and without the financial report parameters, we would like to examine our main experiment question from different aspects, leading to a deeper understanding:

- Per-stock CSVs of actual vs. predicted values in each of the experiments.
- Integration into Streamlit chatbot with Gemini analysis
- Model performance tables and Δ MAPE plots
- Visuals of feature importances and correlation maps-which features had a greater impact on the stock price prediction? When (in which quarters) the biggest impact has been made?